R18

Code No: 152AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, July/August - 2021
BASIC ELECTRICAL ENGINEERING
(Common to EEE, CSE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) State and explain superposition theorem.
b) Find the transient current flowing in the circuit shown in figure 1 if the switch is closed at $t = 0^+$. [7+8]

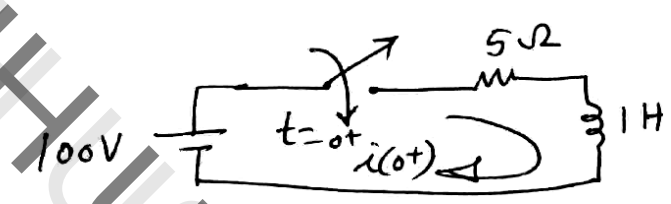


Figure: 1

2.a) How do you find out thevenin's voltage?
b) Find the current 'i' in the circuit shown in figure 2. [6+9]

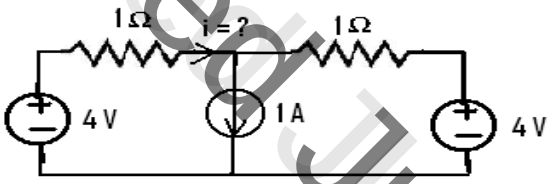


Figure: 2

3.a) Obtain the resonant frequency of R-L-C series circuit excited by sinusoidal voltage source?
b) Find the impedance of the following circuit shown in figure 3. [8+7]

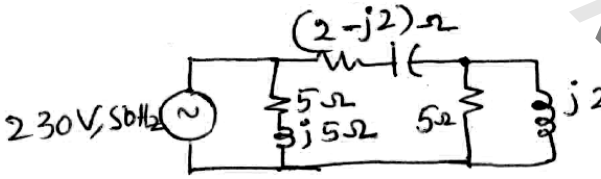


Figure: 3

4.a) Derive the expression for the active power in a series RL circuit excited by sinusoidal source.
b) What is the relationship between line and phase quantities in star connected three phase systems? [7+8]

5. Obtain the equivalent circuit of a transformer. [15]

6.a) What are the applications of auto transformer?
b) Define regulation of a transformer? State different losses of a transformer. [6+9]

7. Explain the working of a synchronous generator. [15]

8. What is the purpose of earthing? Explain any one type of earthing. [15]

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